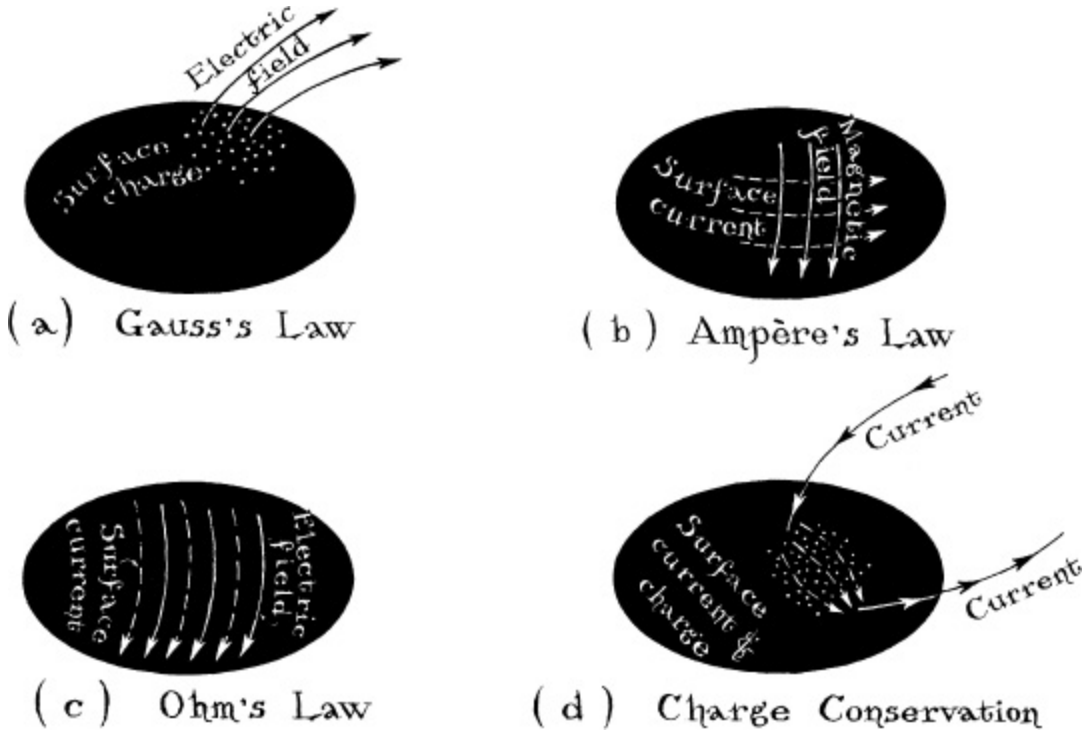
useful on Sunday, when I am puzzling over the pulsations of black holes. The membrane paradigm, with horizons made from electrically charged membranes, may be useful on Monday, when I am puzzling over a black hole's production of jets. And since the predictions of the two paradigms are guaranteed to be the same, I can use each day whichever one best suits my needs.



11.5 The laws governing electric charge and current on a black hole's membrane-like horizon: (a) Gauss's law—the horizon has precisely the right amount of surface charge to terminate all electric field lines which intersect the horizon, so they do not extend into the hole's interior; compare with Figure 11.3. (b) Ampere's law—the horizon has precisely the right amount of surface current to terminate that portion of the magnetic field which is parallel to the horizon, so there is no parallel field below the horizon. (c) Ohm's law—the surface current is proportional to the part of the electric field which is tangential to the surface; the proportionality constant is a resistivity of 377 ohms. (d) Charge conservation—no charge is ever lost or created; all positive charge that enters the horizon from the outside Universe becomes attached to the horizon, and moves around on it, until it exits back into the outside Universe (in the form of negative charge falling inward to neutralize the positive charge).

Not so inside a black hole. Any being who falls into a hole will discover that the horizon is *not* a charge-endowed membrane, and that inside the hole the membrane paradigm completely loses its power. However, infalling beings pay a price to discover this: They cannot publish their discovery in the scientific journals of the outside Universe